LIAM TAYLOR

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Technical Skills

Programming: C, Python (pandas, numpy, astropy,openCV, Mediapipe), MATLAB/Simulink, HTML, JavaScript
Certifications: SAChE Certificates (Process Safety Hazards,Hazard Recognition),CPR, Fire Extinguisher, Respiratory
Software: Altium, Fusion360, VS Code, GitHub, Bootstrap, Microsoft Office (Excel, Word, PowerPoint)

Personal Projects

Copper Electroplating, *Material Science / Electrochemistry Project*

- **Electroplated** a copper coating onto an organic specimen using a conductive graphite-painted cathode in a **copper sulfate** electrolytic cell with a copper-wire anode and 6V DC source, depositing over 10g of copper in 6 hours; applied **Faraday's laws** of **electrolysis** to relate deposited mass to charge and current.

IEEE EXO — Project Lead, *Robotic Gait-Assist Leg System*

- Leading a cross-organizational **robotic exoskeleton** project through **IEEE EXO**, an **electrical engineering student organization**, and a campus research lab, coordinating four subsystem teams against a two-semester timeline and **$5,000 budget**; authored the project's governance and requirements framework to drive accountable, independent parallel workstreams.

Mime, *IEEE EXO Flagship Project — Gesture-Controlled Robotic Hand*

- Developed a dual-mode (MediaPipe webcam vision and flex-sensor glove) gesture control system for a **5-finger tendon-driven robotic hand** on an **ESP32** platform, with calibration tooling and a debounced real-time **gesture interface**.

Court Commander, *Personal Project — Vanilla JavaScript, HTML5 Canvas, SheetJS*

- Built a browser-based volleyball **coaching application** handling live scoring, automatic rotation logic, dual-team lineup management, and a 6-rotation planner with stat-tracking and XLSX export via **SheetJS**, used as the primary in-game tool for a competitive high school program.

Research Experience

Undergraduate Research Assistant, TROPICS Lab, Dept. of Electrical Engineering, USF Sep 2026 – Present

- Working under PhD mentors Mrinmoy Turja and Mitchell Harrah on **Diffuse Correlation Spectroscopy** (DCS) instrumentation, **avalanche photon-counting** paired with **FPGA/correlator data acquisition** for noninvasive deep-tissue blood-flow measurement. Currently assisting in developing an improved sensor design and power system.

Work Experience

Assembly Technician 2, Lithionics, Clearwater, FL Jan 2025 – Jul 2026

- Performed **First Article Inspections** on **400A-rated UL-certified** Battery Management Systems and **12V/24V/51V/102V** single-module lithium-ion batteries, verifying internal data-monitoring components and physical build integrity.
- Conducted **discharge testing** and **final-approval inspections** on multi-module battery systems, maintaining in-house data records against customer specifications and using findings to drive process-checklist improvements.
- Collaborated with engineering to **redesign** a power-connection voltage/polarity testing tool now standard in the production process, **reducing defect rates** and improving technician safety during module testing.

Assistant Foreman, TE Taylor Co., Tampa, FL Jun 2018-Aug 2024

- Direct and coordinate multiple company leaders and crew members in commercial development of **ADA-compliant** concrete sidewalks, ramps, and drainage valleys.
- Evaluate site conditions and project plans with head foreman to optimize workflow, reducing overtime and **increasing completion rates**.

Education

Bachelor of Science in Electrical Engineering, Concentration in Power System Jan 2026-Present
University of South Florida, Tampa, FL
Bachelor of Science in Physical Sciences 2018- Dec 2024
University of South Florida, Tampa, FL
***Relevant coursework* :** Electrical Systems I & II, Digital Circuits, Electronic Materials, Engineering Analysis, Material and Energy Balances, Transport phenomena, Thermodynamics, Organic and Physical Chemistry, Modern Physics, Astrophysics